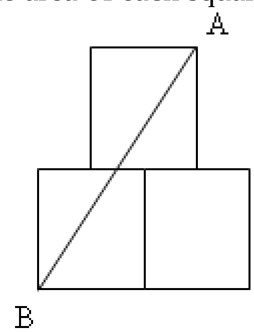
# RMO(ORISSA) – 2004

Max Mark – 100 Time – 3 Hrs

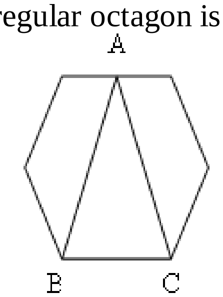
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.
- Q. 1 to Q.2 carry 10 marks each & Q. 3 to Q. 12 carry 8 marks each.
- Bonus question will be taken into account in case of tie.

1.
   (a) Figure below contains 3 squares such that upper square is kept at the midpoints of side of below two squares. If AB = 100 then find the area of each square..




   (b) In the figure what part of Area of regular octagon is the area of triangle ABC?




2.
   (a) For real numbers a, b such that $f(ab) = f(a) + f(b)$ and $f(2) = x, f(5) = y$ then find $f(100)$ in terms of x and y.
   (b) Write in ascending order $2^{48}, \ 3^{36}, \ 5^{24}$.
3.
   (a) Find the remainder when 123456789 divided by 8.
   (b) In a pole of n ft height we have to arrange 4 flags of different color, where red color flag height is 2 ft, white, blue and yellow flags each are of 1 ft. If $a_n$ be the all possible arrangement of flags then what is the relation between $a_n, a_{n-1}, a_{n-2}$.
4. If $a > 0$, then find all positive real roots of $ax - ax^2 = [x] + \frac{1}{2}$, where [x] is the greatest integer less than equals to x.
5. Write the integral part of $\dfrac{1}{\frac{1}{1984} + \frac{1}{1985} + \ldots\ldots\ldots + \frac{1}{1989}}$.

6. Find value of $\sqrt{\underbrace{111...11}_{2000\,digits} - \underbrace{222...22}_{1000\,digits}}$ if it is a natural number.
7. Find all function $f : R \to R$, such that $(x-y)f(x+y)-(x+y)f(x-y)=4xy(x^2-y^2)$.
8. $u+v+w=1$, for all non negative real numbers u, v, w prove that $\sqrt{12uvw}+u^2+v^2+w^2 \le 1$.
9. In triangle ABC, if $BE \perp AC, CF \perp AB$, K is the midpoint of EF and L is midpoint of BC the prove that $m\angle KAE = m\angle BAL$.
10. A, B, C are lies on a straight line. K is a point outside the straight line. If $S_1, S_2, S_3$ are the circum centre of triangle KAB, KAC and KBC respectively, then prove that $K, S_1, S_2, S_3$ lies on a circle.
11. P is an interior point of an equilateral triangle ABC. If $AP^2 = BP^2 + CP^2$, then prove that $m\angle BPC = 150^0$.
12. If $a_1, a_2, ........, a_n$ are n real numbers, then prove that there exist one real number $\alpha$ such that each $a_1+\alpha, a_2+\alpha, ........, a_n+\alpha$ are irrational numbers.

(BONUS QUESTION)

13. If n is an odd natural number, then prove that last two digits of $2^{2n}(2^{2n+1}-1)$ is 28.

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